
2. Introduction

Knowledge Discovery in Databases with Exercises

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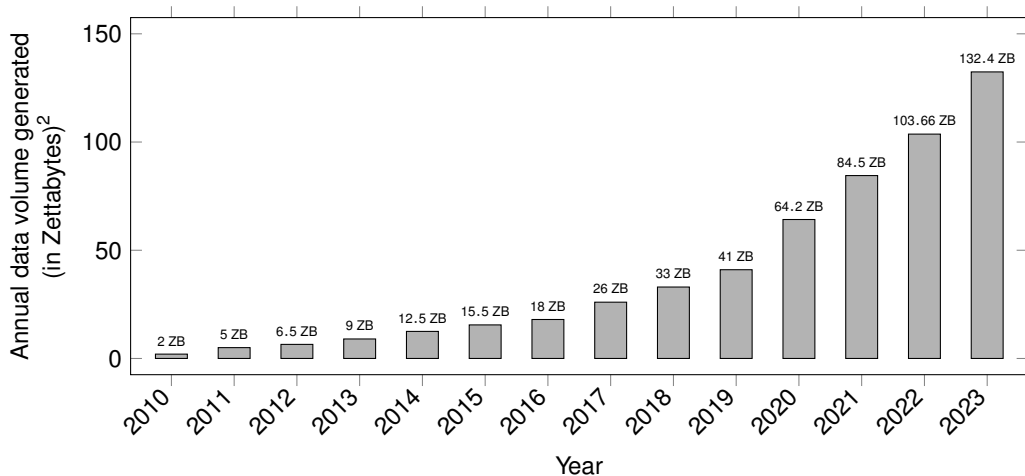
5. Summary

Why Data Mining?

The explosive growth of data: from petabytes to exabytes to zettabytes¹ and beyond.

- Data collection and availability:
 - Automated data collection tools.
 - Database systems.
 - World wide web.
 - Computerized society.
 - Digitization.
- Major sources of abundant data:
 - Business: web, e-commerce, transactions, stocks ...
 - Science: remote sensing, bioinformatics, scientific simulation ...
 - Society: news, digital cameras, social media ...
- The era of **big data** (as inflationary used buzzword).

¹ 1 Zettabyte = 1.000.000.000.000.000.000 Byte (21 Zeros!)



²Source: <https://de.statista.com/statistik/daten/studie/267974/umfrage/prognose-zum-weltweit-generierten-datenvolumen/>

The initial situation:

- We are drowning in data
- We are starving for knowledge

The basic idea behind data mining:

- We can analyze the data to satisfy our hunger for knowledge

What Is Data Mining?

Data mining or knowledge discovery from data:

- Extraction of interesting (**non-trivial, implicit, previously unknown and potentially useful**) patterns from huge amounts of data.
- Is **data mining** a misnomer?

Alternative names:

- Knowledge discovery/mining in databases (KDD).
- Knowledge extraction.
- Data/pattern analysis.
- Data archeology/dredging.
- Information harvesting.
- Business intelligence.

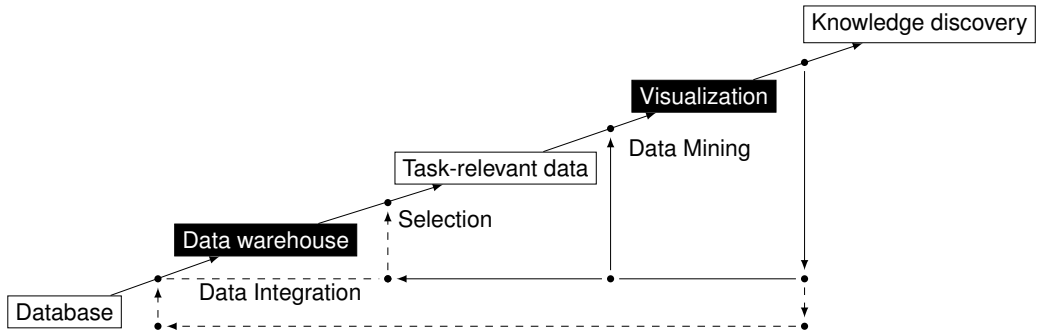
Considered to be data mining:

- Analysis of customer behavior for user-related advertising.
- Analysis of payment histories for fraud detection.
- Analysis of infection behavior for better understanding of a pandemic.

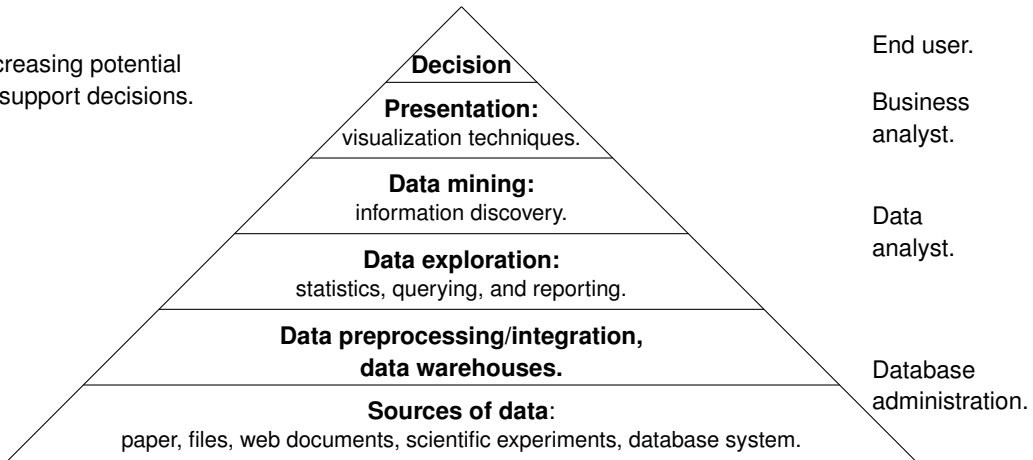
NOT considered to be data mining:

- Simple search for females in a customer database.
- Simple join of two database tables.
- Simple deductive database validating a new tuple with regards to predefined constraints.

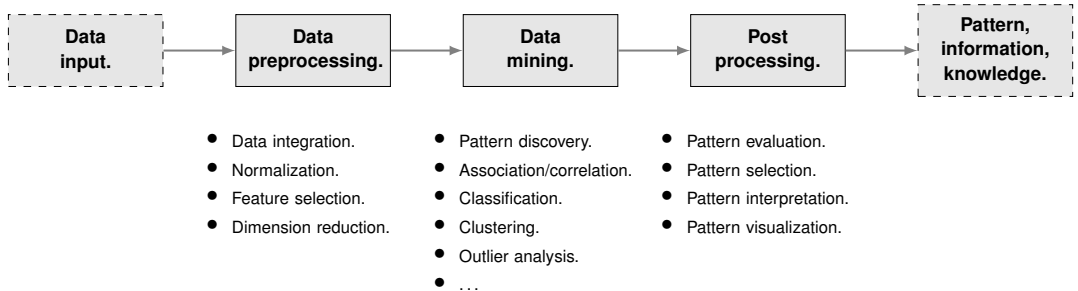
- The **Knowledge discovery pipeline** is a typical view from the database community.
- Data mining plays an essential role in the knowledge-discovery process.



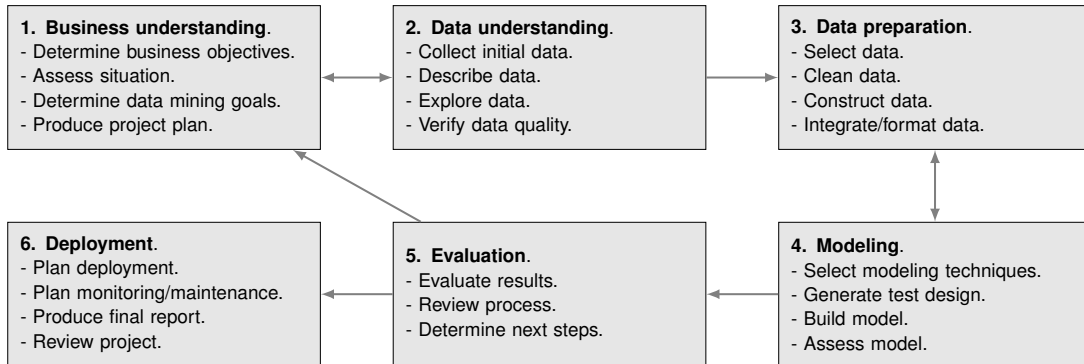
Increasing potential
to support decisions.



The Machine Learning community usually classifies data mining as the central part of its pipeline:



- **CRoss-Industry Standard Process for Data Mining:**



A Multidimensional View of Data-Mining

Data mining projects can be described in four dimensions:

- **What data is available?:**
Data can exist in a wide variety of forms and must therefore be treated differently in data mining.
- **What patterns are searched for?:**
Various functions in data mining can be used to detect different patterns.
- **What technologies are used?:**
The technologies used can vary greatly in data mining.
- **What is the actual target application?:**
The actual target application also differs from case to case.

A Multidimensional View of Data-Mining

What Data Is Available?

- **Any kind of data as long as meaningful for the target application.**
- Most basic forms of data sources:
 - **Relational database:**
Collection of tables, where the tables consist of a set of attributes and usually a large set of tuples.
 - **Data warehouse:**
Repository of information collected from multiple sources, stored under a unified schema.
 - **Transactional database:**
Captures transactions, such as customer purchases, flight bookings, or user clicks on a website.

Advanced data sets and advanced applications:

- Data streams and sensor data.
- Time series data, temporal data, sequence data (incl. biosequences).
- Structure data, graphs, social networks and multi-linked data.
- Object-relational databases.
- Heterogeneous databases and legacy databases.
- NoSQL databases.
- Spatial data and spatiotemporal data.
- Multimedia databases.
- Text databases.
- The world wide web.

A Multidimensional View of Data-Mining

What Patterns Are Searched For?

- **Searching for the right patterns is important.**
- Which patterns can be mined depends on:
 - **Data mining function.**
Different functions can reveal different patterns.
 - **Data set.**
Some types of records contain special patterns that can be found only in them.
- Patterns do not always lead to useful information.
→ Always validate whether the gained knowledge is interesting.

Some will be covered in this lecture:

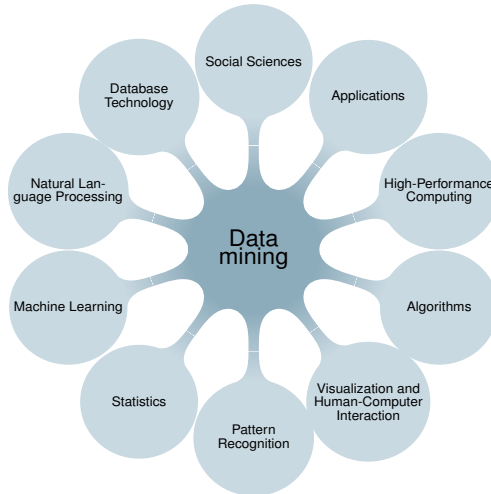
- Data Transformations → *Lectures 3, 4, and 5*
- Frequent Pattern Analysis → *Lecture 6*
- Classification → *Lecture 7*
- Clustering → *Lecture 8*
- Outlier Analysis → *Lecture 9*

But where are many more. E.g.:

- Network Analysis
- Web Mining
- Time Series Analysis
- Neural Networks
- . . .

A Multidimensional View of Data-Mining

What Technologies Are Used?



Each discipline contributes something different. E.g.:

- **Algorithms:**
Basic algorithms to get started.
- **Machine Learning:**
Specialized algorithms for learning from data.
- **High-Performance computing:**
Parallel and distributed computing to handle large datasets.
- **Database Technologies:**
Efficient storage and retrieval of data.
- **Etc.:**
...

A Multidimensional View of Data-Mining

What Is the Actual Target Application?

- **Wherever there is data and more knowledge is desired, there are data mining applications.**
- Typical data mining applications:
 - **Business Intelligence**
Provides historical, current, and predictive views of business operation.
 - **Web Search Engines**
Need to decide which pages to index, which ones to index and how to rank them for search.
 - **Fraud detection**
Possible fraud attempts automatically based on suspicious patterns in transactions.
 - **Predictive Maintenance**
Evaluation of sensor data to maintain machines in time before a defect occurs.

- Example research projects using data mining at FAU³:
 - **Prediction of product properties using data mining methods.**
Prof. Dr.-Ing. Sandro Wartzack (Chair of Engineering Design)
 - **Combustion and fuel optimization for the utilization of residues in biomass furnaces.**
Prof. Dr.-Ing. Jürgen Karl (Chair of Energy Process Engineering)
 - **CoralTrace – A new approach to understanding climate-induced reef crises.**
Prof. Dr. Wolfgang Kießling (Chair of Palaeontology)
 - **Performance Analysis in Team Sports.**
Prof. Dr. Björn Eskofier (Machine Learning and Data Analytics Lab)
 - **And many more.**
Chair of computer science 6 (data management) has some projects related to data mining, too. More information will be given in the last lecture.

³Found in the FAU CRIS (Current research information system): <https://cris.fau.de/>

Major Challenges in Data Mining

Mining methodology:

- Mining various and new kinds of knowledge.
- Mining knowledge in multi-dimensional space.
- Data mining: An interdisciplinary effort.
- Boosting the power of discovery in a networked environment.
- Handling noise, uncertainty, and incompleteness of data.
- Pattern evaluation and pattern- or constraint-guided mining.

User interaction:

- Interactive mining.
- Incorporation of background knowledge.
- Presentation and visualization of data mining results.

Efficiency and scalability:

- Efficiency and scalability of data-mining algorithms.
- Parallel, distributed, stream and incremental mining methods.

Diversity of data types:

- Handling complex types of data.
- Mining dynamic, networked and global data repositories.

Data mining and society:

- Social impacts of data mining.
- Privacy-preserving data mining.
- Invisible data mining.

Summary

In this lecture:

- **Data Mining:**
Discovering interesting patterns and knowledge from massive amounts of data.
- **Data Mining Process:**
Every community focusses on different aspects.
- **Important Aspects:**
What Data? What Patterns? What Technologies? What Applications?
- **Challenges:**
A lot of things to consider.

Any questions about this chapter?

Ask them now or ask them later in our forum:



 https://www.studon.fau.de/studon/goto.php?target=1code_OLYeD79h

Appendix

- Before 1600, era of **empirical science**.
- 1600 — 1950s, rise of **theoretical science**.
 - Each discipline has grown a theoretical component.
 - Theoretical models often motivate experiments and generalize our understanding.
- 1950 — 1990s, rise of **computational science**.
 - Over the last 50 years most disciplines have grown a third, computational branch.
 - E.g. empirical, theoretical, and computational ecology.
 - E.g. physics, linguistics or biology.
 - Computational science traditionally meant simulation.
 - It grew out of our inability to describe reality by closed-form mathematical models.

- 1990—now, rise of **data science**.
 - The flood of data from new instruments and modern simulations.
 - The ability to economically store and manage petabytes of data.
 - The internet makes all these archives world wide accessible.
 - Scientific *information management*,
acquisition,
organization,
query, and
visualization scale almost linearly with amount of data.
 - **Data mining** is a major new challenge!
- For further reading:
Jim Gray and Alex Szaly: *The World Wide Telescope: An Archetype for Online Science*,
Communications of the ACM 45(11): 50-54, 2002.

- 1960s: Data collection, database creation, integrated management systems (IMS), and network database management systems (DBMS).
- 1970s: Relational data model, relational DBMS implementation (RDBMS).
- 1980s: RDBMS products, database creation, advanced data models (extended relational, object oriented, deductive etc.), application-oriented DBMS (spatial, scientific, engineering etc.).
- 1990s: Data mining, data warehousing, multimedia databases, web databases.
- 2000s: Stream data management and mining, data mining and applications, web technology (XML, data integration), and global information systems.

- **1989 IJCAI Workshop on Knowledge Discovery in Databases:**
Knowledge Discovery in Databases (G. Piatetsky-Shapiro and W. Frawley, 1991).
- **1991-1994 Workshops on Knowledge Discovery in Databases:**
Advances in Knowledge Discovery and Data Mining (U. Fayyad, G. Piatetsky-Shapiro, P. Smyth and R. Uthurusamy, 1996).
- **1995-1998 International Conferences on Knowledge Discovery in Databases and Data Mining (KDD'95-98):**
Journal of Data Mining and Knowledge Discovery (1997).
- **ACM SIGKDD conferences since 1998 and SIGKDD Explorations.**
- **More conferences on data mining:**
PAKDD (1997), PKDD (1997), SIAM-Data Mining (2001), (IEEE) ICDM (2001), etc.
- **Journal ACM Transactions on KDD starting in 2007.**

KDD Conferences:

- ACM SIGKDD Int. Conf. on Knowledge Discovery in Databases and Data Mining (KDD).
- SIAM Data Mining Conf. (SDM).
- (IEEE) Int. Conf. on Data Mining (ICDM).
- European Conf. on Machine Learning and Principles and Practices of Knowledge Discovery and Data Mining (ECML-PKDD).
- Pacific-Asia Conf. on Knowledge Discovery and Data Mining (PAKDD).
- Int. Conf. on Web Search and Data Mining (WSDM).

Other related conferences:

- DB conferences: ACM SIGMOD, VLDB, ICDE, EDBT, ICDT, ...
- Web and IR conferences: WWW, SIGIR, WSDM, ...
- ML conferences: ICML, NIPS, ICLR ...
- PR conferences: CVPR, ICPR ...

Journals:

- Data Mining and Knowledge Discovery (DAMI or DMKD).
- IEEE Trans. On Knowledge and Data Eng. (TKDE).
- KDD Explorations.
- ACM Trans. on KDD.

Data mining and KDD (SIGKDD: CD-ROM):

- Conferences: ACM-SIGKDD, IEEE-ICDM, SIAM-DM, PKDD, PAKDD, etc.
- Journal: Data Mining and Knowledge Discovery, KDD Explorations, ACM TKDD.
- KDnuggets: www.kdnuggets.com.

Database systems (SIGMOD: ACM SIGMOD Anthology CD-ROM):

- Conferences: ACM-SIGMOD, ACM-PODS, VLDB, IEEE-ICDE, EDBT, ICDT, DASFAA.
- Journals: IEEE-TKDE, ACM-TODS/TOIS, JIIS, J. ACM, VLDB J., Info. Sys., etc.

AI & Machine Learning:

- Conferences: Machine learning (ML), AAAI, IJCAI, COLT (Learning Theory), CVPR, NIPS, etc.
- Journals: Machine Learning, Artificial Intelligence, Knowledge and Information Systems, IEEE-PAMI, etc.

Web and IR:

- Conferences: SIGIR, WWW, CIKM, etc.
- Journals: WWW: Internet and Web Information Systems.

Statistics:

- Conferences: Joint Stat. Meeting, etc.
- Journals: Annals of Statistics, etc.

Visualization:

- Conferences: CHI, ACM-SIGGraph, etc.
- Journals: IEEE Trans. Visualization and Computer Graphics, etc.